**Supraventricular Tachycardia: Adult and Pediatric**
**C-8 Effective: June 10, 2024**

Adult symptomatic tachycardia usually occurs at rates of ≧ 150 and has a narrow complex, is regular, and has no discernible p waves.

Most important goal is to differentiate if STABLE or UNSTABLE. A STABLE adult patient may complain of palpitations, chest discomfort, respiratory distress. UNSTABLE adult patient may experience one or more of the following: B/P < 90 with signs of shock / poor perfusion, or acutely decreasing level of consciousness.

Typical adult sinus tachycardia is in the range of 100 to (220 - patient's age) beats per minute. For any tachycardia with rate approximately 100-150 and a normal blood pressure, consider CLOSE OBSERVATION and/or fluid bolus rather than immediate treatment with an anti-arrhythmic medication. Search for underlying cause of tachycardia such as fever, sepsis, dyspnea, etc.

If patient has history or 12 Lead ECG reveals Wolf Parkinson White (WPW), use caution with Adenosine and give only with defibrillator available.

Using adenosine in a patient with a history of asthma or COPD may precipitate a bronchospasm.

If the QRS complexes are wider than 0.12 seconds, consider Ventricular Tachycardia. Refer to that protocol.

In the event of cocaine, methamphetamine, TCA intoxication, refer to the Suspected Overdose Protocol.

Supraventricular tachycardia (SVT) in the pediatric patient is regular, at a rate ≧ 220 in the infant and ≧ 180 in the child. Sinus Tachycardia in the child or infant associated with poor perfusion should be treated in accordance with the Hypotension Shock protocol.

Hypotension can be defined as a systolic blood pressure below the 5th percentile in a child/infant.

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| Infant less than 1 month | 60 mmHg systolic |
| 1 month to 1 year | 70 mmHg systolic |
| 1 year to 10 years | 70 plus 2 X age in years systolic |
| ➢ 10 years | 90 mm Hg systolic |

Begin Initial Medical Care.

Obtain and transmit 12 lead ECG if possible.

STABLE ADULT PATIENT:

- Try Valsalva.
- IF RHYTHM IS NOT ATRIAL FIBRILLATION administer **adenosine 12 mg** rapid IV push over 1-3 seconds to be followed by a 20 ml saline flush. If no response, **may repeat in 1-2 minutes** at 12 mg IV push over 1-3 seconds followed by 20 ml saline flush.

- If rhythm is an irregular and narrow complex tachycardia that suggest atrial fibrillation with a rapid ventricular response and patient presents with signs and symptoms of hypoperfusion:
    1. If signs and symptoms suggest hypovolemia, administer 500ml of IV fluid.
    2. If signs and symptoms suggest cardiogenic etiology, administer Diltiazem 0.25mg/kg over 2 minutes via IV/IO.

STABLE PEDIATRIC PATIENT: Patient is alert, with minimal signs of poor perfusion, and is NOT hypotensive.

Consider vagal maneuvers to include application of iced glove or ice pack to face of infant and blowing into an occluded straw device for the child.

**Adenosine 0.1 mg/Kg** IV "AT MD DISCRETION ONLY"

UNSTABLE ADULT PATIENT: HR ≧ 150 with a decreased LOC and/or B/P < 90 systolic with signs of shock/poor perfusion.

**Adult Pain Management for Synchronized Cardioversion:**
***DO NOT DELAY LIFE SAVING CARE IN ORDER TO SEDATE THE SEVERELY COMPROMISED PATIENT*.**
**If sedation needed:**
**Versed: 10** mg midazolam IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP if:

GCS ≥ 14

ATTEMPT SYNCHRONIZED CARDIOVERSION WITHIN THE FOLLOWING RANGES:

50-100 J, 200 J, 300 J, 360 J if arrhythmia is SVT
125 J, 200 J, 300 J, 360 J if arrhythmia is suspected atrial fibrillation.

**Assess LOC, pulses and B/P between each shock delivery to confirm need to escalate dosing**

UNSTABLE PEDIATRIC PATIENT: Patient demonstrates a decreased LOC, poor perfusion and hypotensive as defined by a blood pressure below the 5th percentile.

- **If IV or IO is readily available**. Administer **Adenosine 0.1 mg/kg (Max dose 6mg)** rapid IV push followed by normal saline flush of 2-5 mL. If ineffective Adenosine may be attempted again once with an increased dose of 0.2 mg/kg (max dose 12mg).
- Confirm all doses with the length-based tape.

- **If IV or IO is not readily available**, synchronize cardiovert at 0.5-1 J/kg, repeated at 2 J/kg if needed.
- Pediatric Pain Management for Synchronized Cardioversion is at MD Discretion only.